Generalized Additive Model for Council District IV (2023 - 2025 Sales)

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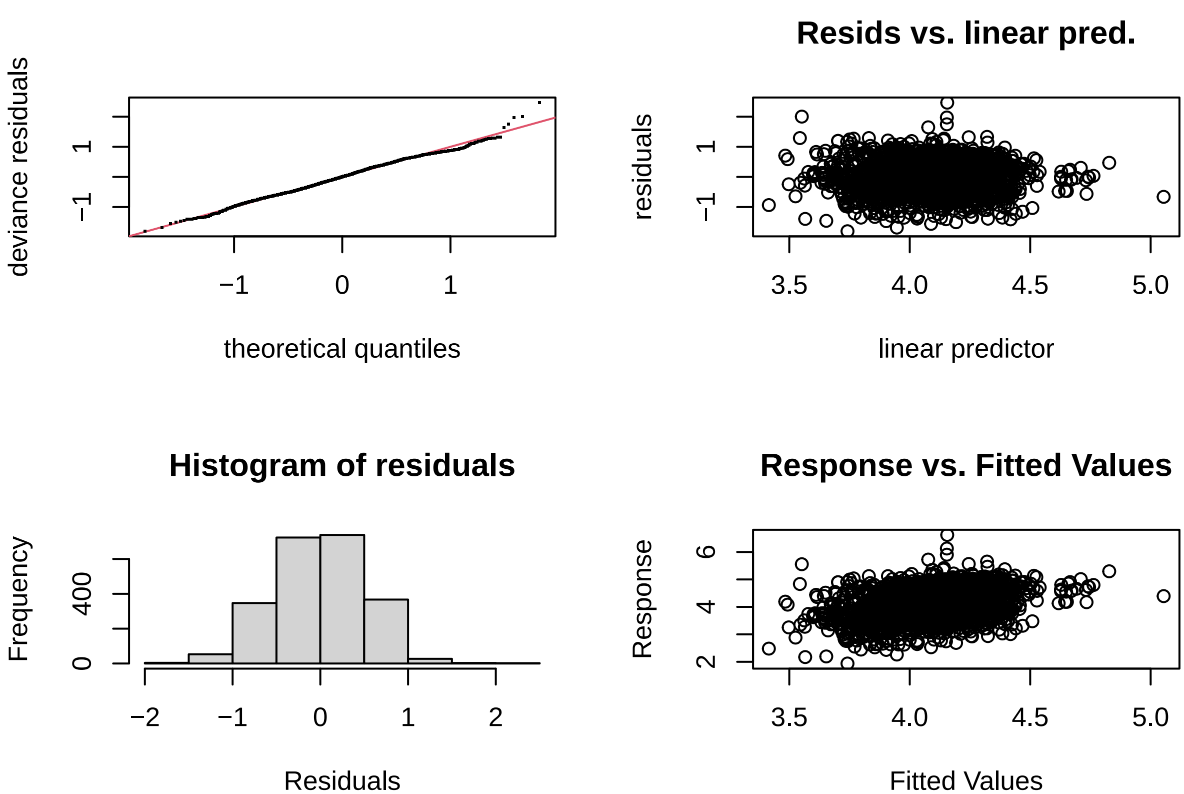
Contents

Model I

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_sppsf ~ arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          4.070996   0.038278 106.354  < 2e-16 ***
## arterial             0.007219   0.039027   0.185 0.853266
## single_family1story -0.050203   0.032208  -1.559 0.119208
## half_duplex1story    0.043962   0.084520   0.520 0.603020
## half_duplex2story    0.139141   0.114502   1.215 0.224423
## smallest             0.260355   0.048863   5.328 1.09e-07 ***
## small                0.118451   0.036863   3.213 0.001331 **
## large               -0.041725   0.033338  -1.252 0.210853
## largest             -0.137245   0.063073  -2.176 0.029663 *
## two_plus_baths       0.160938   0.099052   1.625 0.104351
## powder_room          0.043820   0.045698   0.959 0.337713
## good                 0.469917   0.100134   4.693 2.86e-06 ***
## fair                -0.182547   0.027512  -6.635 4.05e-11 ***
## poor                -0.320112   0.067999  -4.708 2.66e-06 ***
```

```
## very_poor_unsound   -0.396423   0.117600  -3.371 0.000762 ***
## qabove_avg           0.362476   0.143157   2.532 0.011409 *
## qbelow_avg          -0.068556   0.032204  -2.129 0.033380 *
## crawl               -0.007714   0.077434  -0.100 0.920654
## slab                -0.094073   0.190803  -0.493 0.622035
## fireplace            0.139127   0.026562   5.238 1.78e-07 ***
## hot_water            0.105489   0.043474   2.426 0.015325 *
## elec_baseboard      -0.450809   0.545556  -0.826 0.408706
## wall                -0.096124   0.178329  -0.539 0.589925
## central_air          0.124552   0.030853   4.037 5.60e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                       edf Ref.df     F  p-value
## s(sresb_yearbuilt) 4.619      9 4.280  < 2e-16 ***
## s(ln_garagespaces) 1.415      3 4.886 0.000117 ***
## s(months)          1.849      9 2.978 3.97e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.113   Deviance explained = 12.5%
## -REML = 1778.7  Scale est. = 0.2696    n = 2264
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-9.862397e-07,2.51568e-09]
## (score 1778.749 & scale 0.2695985).
## Hessian positive definite, eigenvalue range [0.4715387,1120.006].
```

```
## Model rank =  45 / 45
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                      k'  edf k-index p-value
## s(sresb_yearbuilt) 9.00 4.62    0.89  <2e-16 ***
## s(ln_garagespaces) 3.00 1.42    0.85  <2e-16 ***
## s(months)          9.00 1.85    0.91  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod   prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl> <dbl>
## 1  2264     0.000773   0.000911    0.000658  1.38  51.8 -1.07
```

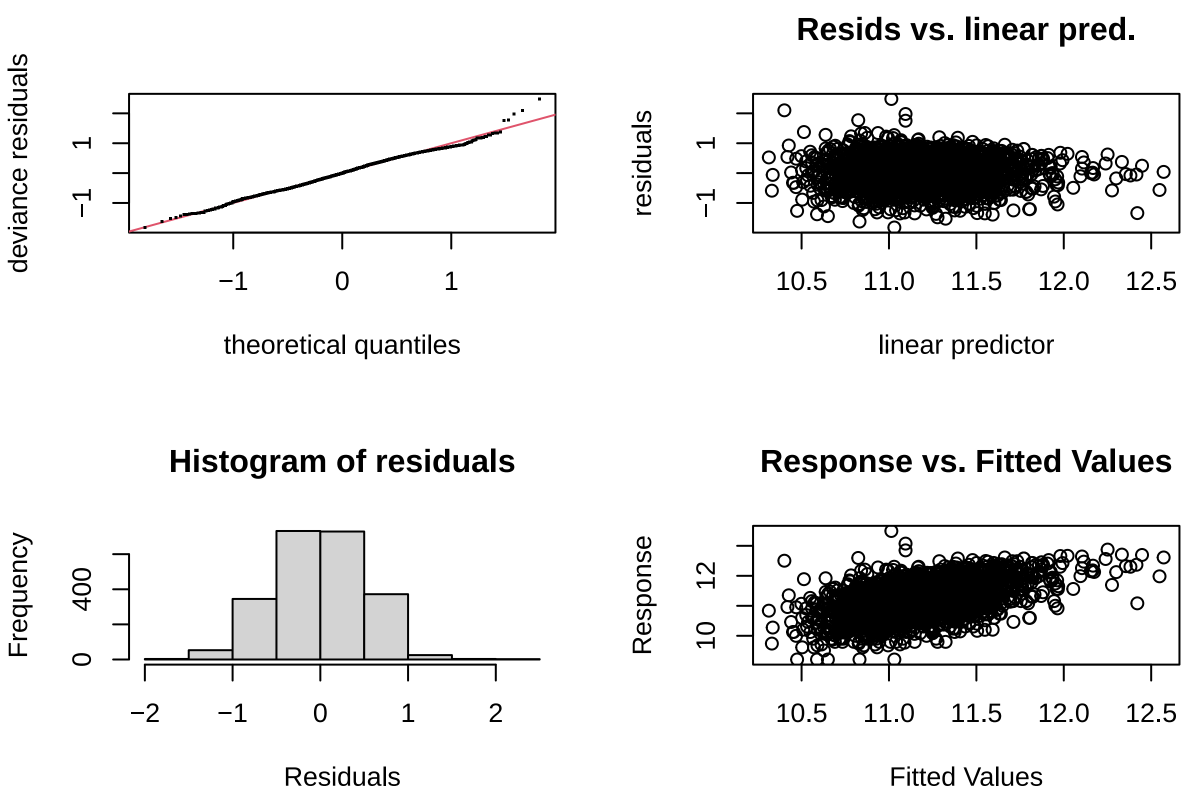
Table 1: District 4 LN_SPPSF NLM I Assessing Coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2264	0.0008457	0.000996	0.0007201	1.383208	51.4283	-0.0008955

Model II

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         11.169122   0.041042 272.139  < 2e-16 ***
## arterial             0.035573   0.039259   0.906  0.36498
## single_family1story -0.052209   0.032322  -1.615  0.10639
## half_duplex1story   -0.222821   0.091465  -2.436  0.01492 *
## half_duplex2story   -0.106292   0.117479  -0.905  0.36568
## smallest             0.254909   0.097040   2.627  0.00868 **
## small                0.121445   0.049472   2.455  0.01417 *
## large               -0.060663   0.048488  -1.251  0.21103
## largest             -0.152292   0.090384  -1.685  0.09214 .
## two_plus_baths       0.156587   0.119525   1.310  0.19031
## powder_room          0.038935   0.050341   0.773  0.43935
```

```
## good                    0.459631   0.099919   4.600 4.46e-06 ***
## fair                   -0.179289   0.027339  -6.558 6.76e-11 ***
## poor                   -0.308167   0.067568  -4.561 5.37e-06 ***
## very_poor_unsound      -0.375877   0.116829  -3.217  0.00131 **
## qabove_avg              0.328698   0.142347   2.309  0.02103 *
## qbelow_avg             -0.055601   0.032790  -1.696  0.09009 .
## crawl                  -0.008293   0.077214  -0.107  0.91448
## slab                   -0.118500   0.195079  -0.607  0.54362
## fireplace               0.120756   0.026947   4.481 7.79e-06 ***
## hot_water               0.094226   0.043301   2.176  0.02966 *
## elec_baseboard         -0.470728   0.545351  -0.863  0.38814
## wall                   -0.179827   0.178433  -1.008  0.31365
## central_air             0.121653   0.030648   3.969 7.43e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                              edf Ref.df     F  p-value
## s(ln_landratio)            1.789      9 3.687  < 2e-16 ***
## s(ln_base_bldgsize_ratio) 1.973      9 3.470  < 2e-16 ***
## s(sresb_yearbuilt)         4.644      9 3.694 9.78e-07 ***
## s(ln_garagespaces)         1.604      3 4.373 0.000414 ***
## s(months)                  1.907      9 2.983 4.84e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.257   Deviance explained = 26.9%
## -REML =   1767  Scale est. = 0.26559   n = 2264
```




```
##
```

```
## Method: REML   Optimizer: outer newton
## full convergence after 8 iterations.
## Gradient range [-7.189403e-06,1.136041e-05]
## (score 1766.993 & scale 0.2655872).
## Hessian positive definite, eigenvalue range [0.3930552,1120.008].
## Model rank =  63 / 63
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                              k'  edf k-index p-value
## s(ln_landratio)            9.00 1.79    0.90  <2e-16 ***
## s(ln_base_bldgsize_ratio) 9.00 1.97    0.97    0.11
## s(sresb_yearbuilt)        9.00 4.64    0.89  <2e-16 ***
## s(ln_garagespaces)        3.00 1.60    0.86  <2e-16 ***
## s(months)                 9.00 1.91    0.91  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2264        0.909       1.04       0.804  1.30  46.0 -0.907
```

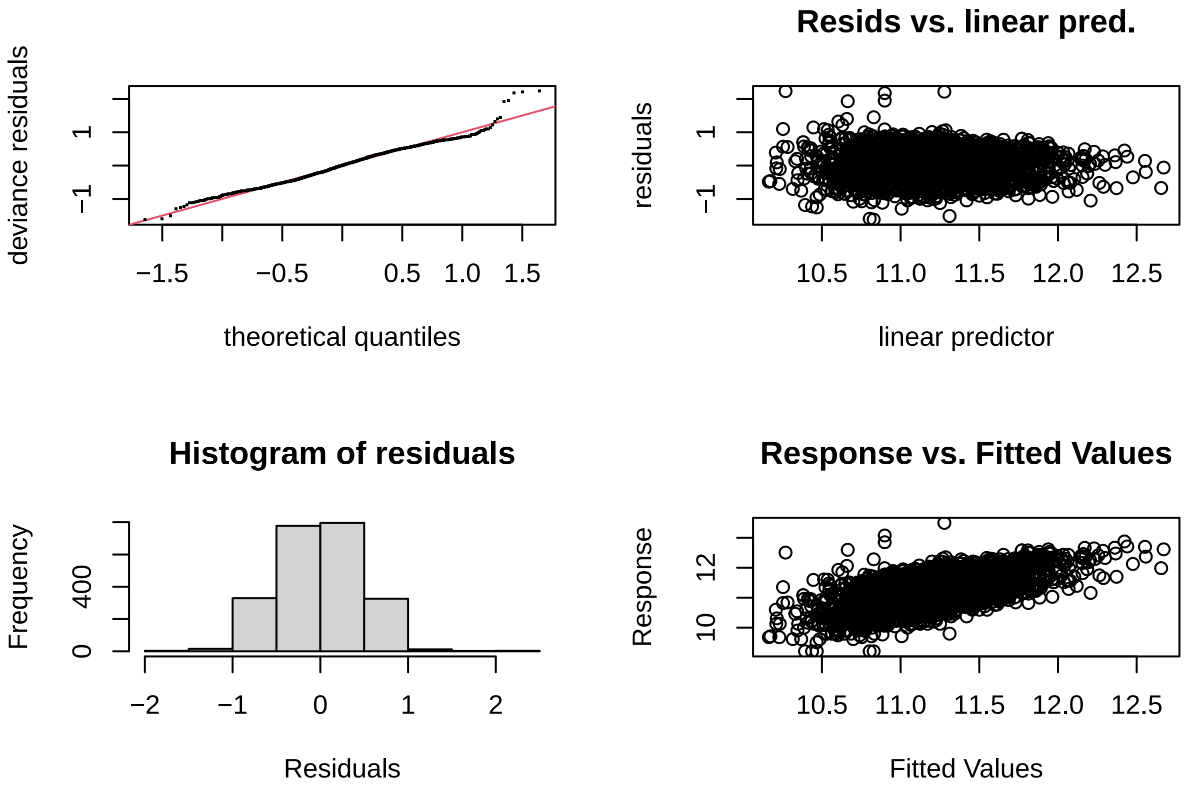
Table 2: District 4 LN_PRICE (No ECFs) NLM I Assessing Coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2264	1.000469	1.139837	0.879581	1.295886	45.37819	-0.8418097

Model III

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10) + ecf
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         10.995230   0.055477 198.193  < 2e-16 ***
## arterial             0.078002   0.036482   2.138 0.032620 *
## single_family1story -0.052731   0.029647  -1.779 0.075438 .
## half_duplex1story   -0.268868   0.084106  -3.197 0.001409 **
```

```
## half_duplex2story   -0.140579   0.108402  -1.297 0.194827
## smallest             0.126057   0.088574   1.423 0.154826
## small                0.054184   0.045336   1.195 0.232145
## large               -0.034407   0.044105  -0.780 0.435414
## largest             -0.041175   0.082211  -0.501 0.616528
## two_plus_baths       0.185163   0.108851   1.701 0.089071 .
## powder_room          0.016010   0.046058   0.348 0.728169
## good                 0.285016   0.090274   3.157 0.001614 **
## fair                -0.177391   0.024981  -7.101 1.66e-12 ***
## poor                -0.259175   0.061728  -4.199 2.79e-05 ***
## very_poor_unsound   -0.231843   0.106664  -2.174 0.029843 *
## qabove_avg           0.106075   0.133082   0.797 0.425497
## qbelow_avg          -0.050985   0.031631  -1.612 0.107139
## crawl               -0.069188   0.071425  -0.969 0.332807
## slab                -0.113867   0.170926  -0.666 0.505366
## fireplace            0.066668   0.024338   2.739 0.006207 **
## hot_water            0.006303   0.038162   0.165 0.868828
## elec_baseboard      -0.142525   0.495787  -0.287 0.773778
## wall                -0.206703   0.163704  -1.263 0.206843
## central_air          0.083575   0.027905   2.995 0.002775 **
## ecf4R402            -0.404090   0.085633  -4.719 2.52e-06 ***
## ecf4R403             0.077860   0.069767   1.116 0.264543
## ecf4R404             0.156427   0.068619   2.280 0.022722 *
## ecf4R405             0.238015   0.055434   4.294 1.83e-05 ***
## ecf4R406             0.190155   0.055084   3.452 0.000567 ***
## ecf4R407             0.058253   0.079329   0.734 0.462831
## ecf4R408             0.072046   0.058215   1.238 0.216001
## ecf4R409             0.163562   0.075977   2.153 0.031443 *
## ecf4R410             0.482093   0.053245   9.054  < 2e-16 ***
## ecf4R411            -0.033942   0.056844  -0.597 0.550497
## ecf4R412            -0.066017   0.079819  -0.827 0.408275
## ecf4R413             0.286292   0.058160   4.922 9.18e-07 ***
## ecf4R414             0.623483   0.072335   8.619  < 2e-16 ***
## ecf4R415             0.779738   0.080697   9.662  < 2e-16 ***
## ecf4R416            -0.184070   0.111904  -1.645 0.100136
## ecf4R417             0.449105   0.058636   7.659 2.78e-14 ***
## ecf4R418             0.101321   0.145806   0.695 0.487191
## ecf4R419            -0.187043   0.078003  -2.398 0.016573 *
## ecf4R421            -0.162373   0.095167  -1.706 0.088113 .
## ecf4R422             0.658480   0.071998   9.146  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                            edf Ref.df     F  p-value
## s(ln_landratio)          0.9063      9 0.418  0.02879 *
## s(ln_base_bldgsize_ratio) 1.9230      9 2.247 7.57e-06 ***
## s(sresb_yearbuilt)       0.5371      9 0.084  0.23838
## s(ln_garagespaces)       1.2281      3 2.579  0.00421 **
## s(months)                2.2673      9 4.448  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.387   Deviance explained = 40.1%
## -REML = 1567.5  Scale est. = 0.21919   n = 2264
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-0.0003420029,0.0002597649]
## (score 1567.526 & scale 0.2191924).
## Hessian positive definite, eigenvalue range [0.07759613,1110.002].
## Model rank =  83 / 83
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                              k'   edf k-index p-value
## s(ln_landratio)           9.000 0.906    1.00    0.42
## s(ln_base_bldgsize_ratio) 9.000 1.923    1.02    0.78
## s(sresb_yearbuilt)        9.000 0.537    0.99    0.32
## s(ln_garagespaces)        3.000 1.228    0.97    0.06 .
## s(months)                 9.000 2.267    0.99    0.31
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

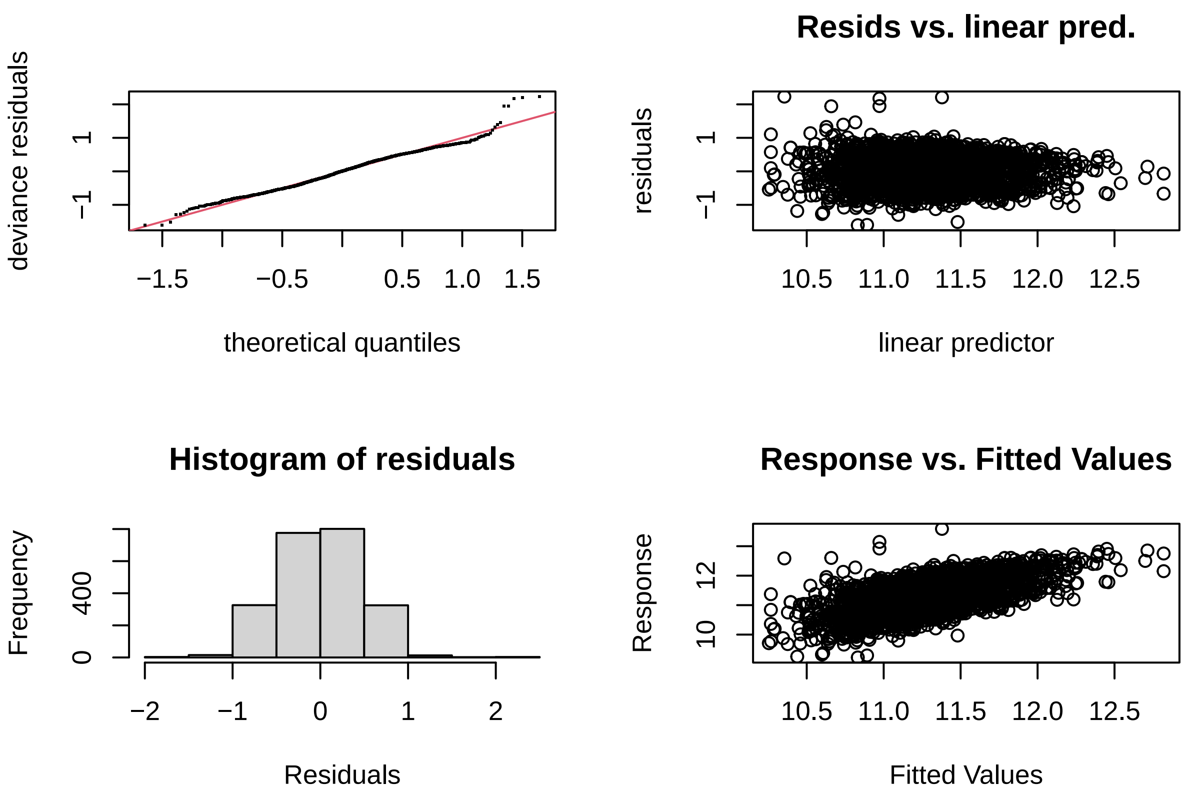
```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2264        0.909       1.02       0.825  1.23  41.2 -0.619
```

Model IV

```
##
```

```
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_tasp ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + ecf + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         11.085989   0.055037 201.427  < 2e-16 ***
## arterial             0.078818   0.036454   2.162 0.030715 *
## single_family1story -0.052714   0.028373  -1.858 0.063314 .
## half_duplex1story   -0.269984   0.083254  -3.243 0.001201 **
## half_duplex2story   -0.141439   0.108311  -1.306 0.191735
## smallest             0.125038   0.088318   1.416 0.156984
## small                0.054088   0.045253   1.195 0.232121
## large               -0.034470   0.044002  -0.783 0.433493
## largest             -0.040083   0.082085  -0.488 0.625375
## two_plus_baths       0.183291   0.108683   1.686 0.091845 .
## powder_room          0.016462   0.045994   0.358 0.720433
## good                 0.281008   0.090165   3.117 0.001853 **
## fair                -0.176192   0.024911  -7.073 2.03e-12 ***
## poor                -0.255853   0.061572  -4.155 3.37e-05 ***
## very_poor_unsound   -0.228703   0.106477  -2.148 0.031829 *
## qabove_avg           0.109507   0.133008   0.823 0.410419
## qbelow_avg          -0.052245   0.031602  -1.653 0.098427 .
## crawl               -0.072442   0.071340  -1.015 0.310003
## slab                -0.107067   0.170642  -0.627 0.530436
## fireplace            0.065833   0.024314   2.708 0.006829 **
## hot_water            0.006592   0.038140   0.173 0.862788
## elec_baseboard      -0.149963   0.495549  -0.303 0.762208
## wall                -0.211306   0.163593  -1.292 0.196610
## central_air          0.083511   0.027889   2.994 0.002780 **
## ecf4R402            -0.400381   0.085547  -4.680 3.04e-06 ***
## ecf4R403             0.078646   0.069731   1.128 0.259505
## ecf4R404             0.156360   0.068617   2.279 0.022779 *
## ecf4R405             0.238980   0.055420   4.312 1.69e-05 ***
## ecf4R406             0.191688   0.055062   3.481 0.000509 ***
## ecf4R407             0.058758   0.079294   0.741 0.458763
## ecf4R408             0.072111   0.058201   1.239 0.215483
## ecf4R409             0.164962   0.075913   2.173 0.029884 *
## ecf4R410             0.482929   0.053197   9.078  < 2e-16 ***
## ecf4R411            -0.032128   0.056794  -0.566 0.571668
## ecf4R412            -0.065263   0.079807  -0.818 0.413584
## ecf4R413             0.287999   0.058136   4.954 7.83e-07 ***
## ecf4R414             0.624238   0.072288   8.635  < 2e-16 ***
## ecf4R415             0.779050   0.080686   9.655  < 2e-16 ***
## ecf4R416            -0.178981   0.111776  -1.601 0.109465
## ecf4R417             0.450917   0.058588   7.696 2.09e-14 ***
## ecf4R418             0.104802   0.145752   0.719 0.472190
## ecf4R419            -0.185125   0.077922  -2.376 0.017597 *
## ecf4R421            -0.157109   0.094642  -1.660 0.097048 .
## ecf4R422             0.659115   0.071980   9.157  < 2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                                 edf Ref.df     F  p-value
## s(ln_landratio)            0.900488      9 0.410  0.03008 *
## s(ln_base_bldgsize_ratio) 1.902350      9 2.223 8.28e-06 ***
## s(sresb_yearbuilt)         0.507369      9 0.078  0.24512
## s(ln_garagespaces)         1.249475      3 2.705  0.00351 **
## s(months)                  0.003026      9 0.000  0.50201
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.383   Deviance explained = 39.6%
## -REML = 1564.6  Scale est. = 0.21921   n = 2264
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 8 iterations.
## Gradient range [-0.0007453549,0.0007591715]
## (score 1564.626 & scale 0.2192106).
## Hessian positive definite, eigenvalue range [0.0007449807,1110.001].
## Model rank =  83 / 83
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                                 k'     edf k-index p-value
## s(ln_landratio)            9.00000 0.90049    1.00    0.46
## s(ln_base_bldgsize_ratio) 9.00000 1.90235    1.02    0.80
```

```
## s(sresb_yearbuilt)        9.00000 0.50737    0.99    0.33
## s(ln_garagespaces)        3.00000 1.24947    0.98    0.10
## s(months)                 9.00000 0.00303    0.99    0.29

## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2264        0.994       1.11       0.903  1.23  40.8 -0.616
```